

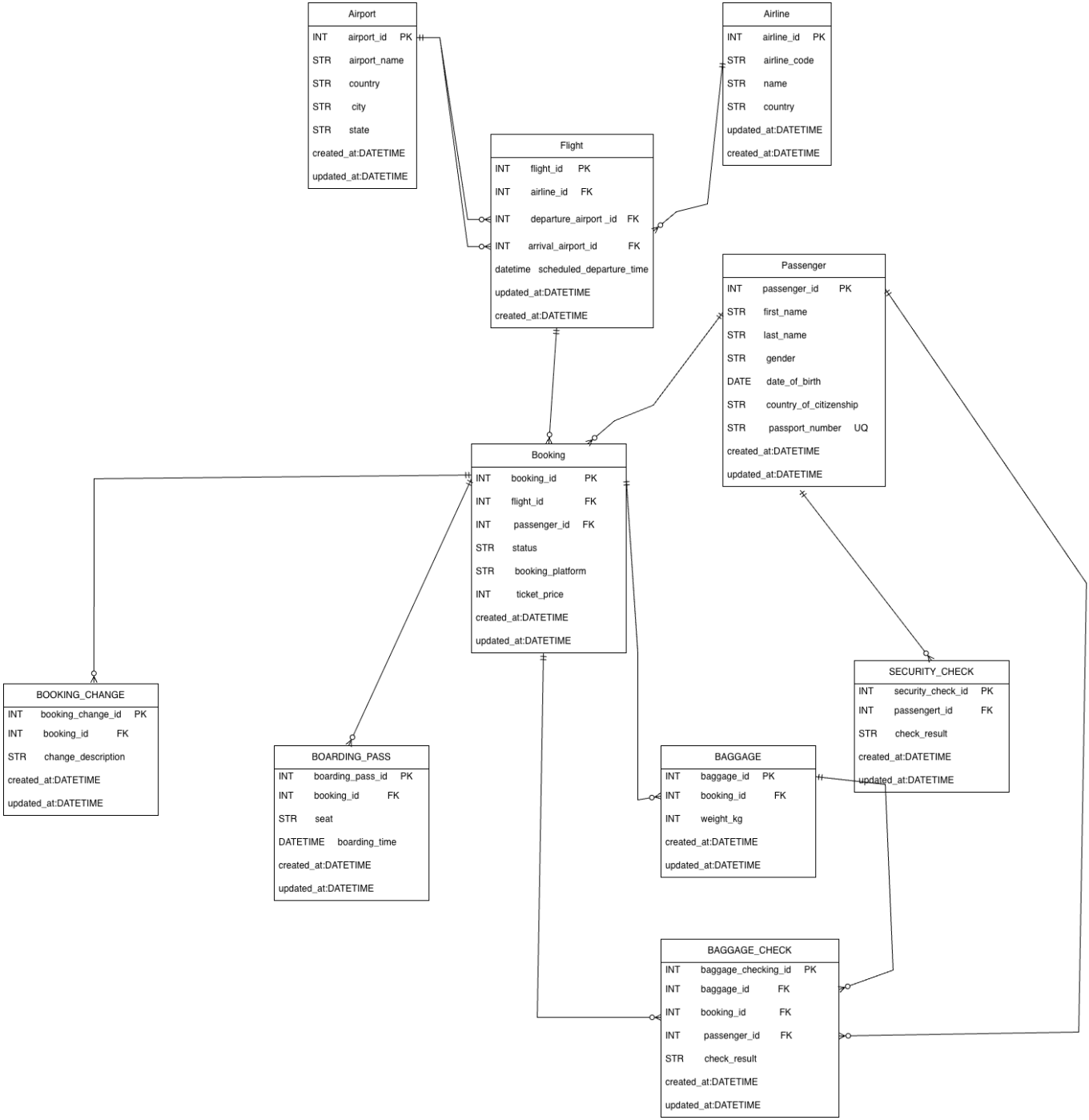
ER Diagram of the Airport Booking System

Report: Entities, Attributes, Relationships and Notation

Bakytzhan Zamanbek

1. ER Diagram

The diagram below shows the entity-relationship (ER) diagram built for the airport booking system. It is drawn using Crow's Foot notation (see the "Notation / Legend" section at the end of this document).



2. Identifying the Entities

Based on the system description and the diagram, the following 10 entities were identified:

Entity	Purpose
Airport	An airport - the departure or arrival point of a flight.
Airline	An airline - the carrier operating a flight.
Flight	A flight - a specific scheduled trip operated by an airline between two airports.
Passenger	A passenger - the individual travelling on a flight.
Booking	A booking - a reservation made by a passenger for a specific flight.
Booking_Change	A booking change - the history of modifications made to a booking.
Boarding_Pass	A boarding pass - the document issued for boarding, tied to a booking.
Baggage	Baggage - a piece of luggage registered under a booking.
Baggage_Check	A baggage check - the result of inspecting a specific piece of baggage.
Security_Check	A security check - the result of a passenger's security screening.

3. Entity Attributes

For each entity, the table below lists its attributes, their data types and constraints: PK - Primary Key, FK - Foreign Key, UQ - Unique value, NN - Not Null (required field).

3.1 Airport

Attribute	Data Type	Constraint
airport_id	INT	PK, NN
airport_name	STR	NN
country	STR	NN
city	STR	NN
state	STR	—
created_at	DATETIME	NN
updated_at	DATETIME	NN

3.2 Airline

Attribute	Data Type	Constraint
airline_id	INT	PK, NN
airline_code	STR	NN, UQ
name	STR	NN
country	STR	NN
updated_at	DATETIME	NN
created_at	DATETIME	NN

3.3 Flight

Flight has two foreign keys referencing Airport: the departure airport and the arrival airport.

Attribute	Data Type	Constraint
flight_id	INT	PK, NN
airline_id	INT	FK -> Airline.airline_id, NN
departure_airport_id	INT	FK -> Airport.airport_id, NN
arrival_airport_id	INT	FK -> Airport.airport_id, NN
scheduled_departure_time	DATETIME	NN
updated_at	DATETIME	NN
created_at	DATETIME	NN

3.4 Passenger

Attribute	Data Type	Constraint
passenger_id	INT	PK, NN
first_name	STR	NN
last_name	STR	NN
gender	STR	—
date_of_birth	DATE	NN
country_of_citizenship	STR	NN
passport_number	STR	UQ, NN
created_at	DATETIME	NN
updated_at	DATETIME	NN

3.5 Booking

Attribute	Data Type	Constraint
booking_id	INT	PK, NN
flight_id	INT	FK -> Flight.flight_id, NN
passenger_id	INT	FK -> Passenger.passenger_id, NN
status	STR	NN
booking_platform	STR	NN
ticket_price	INT	NN
created_at	DATETIME	NN
updated_at	DATETIME	NN

3.6 Booking_Change

Attribute	Data Type	Constraint
booking_change_id	INT	PK, NN
booking_id	INT	FK -> Booking.booking_id, NN
change_description	STR	NN
created_at	DATETIME	NN
updated_at	DATETIME	NN

3.7 Boarding_Pass

Attribute	Data Type	Constraint
boarding_pass_id	INT	PK, NN
booking_id	INT	FK -> Booking.booking_id, NN
seat	STR	NN
boarding_time	DATETIME	NN
created_at	DATETIME	NN
updated_at	DATETIME	NN

3.8 Baggage

Attribute	Data Type	Constraint
baggage_id	INT	PK, NN
booking_id	INT	FK -> Booking.booking_id, NN
weight_kg	INT	NN
created_at	DATETIME	NN
updated_at	DATETIME	NN

3.9 Baggage_Check

This entity references three other tables - Baggage, Booking and Passenger - to record whose baggage it is, under which booking, and the result of the check.

Attribute	Data Type	Constraint
baggage_checking_id	INT	PK, NN
baggage_id	INT	FK -> Baggage.baggage_id, NN
booking_id	INT	FK -> Booking.booking_id, NN
passenger_id	INT	FK -> Passenger.passenger_id, NN
check_result	STR	NN
created_at	DATETIME	NN
updated_at	DATETIME	NN

3.10 Security_Check

Attribute	Data Type	Constraint
security_check_id	INT	PK, NN
passenger_id	INT	FK -> Passenger.passenger_id, NN
check_result	STR	NN
created_at	DATETIME	NN
updated_at	DATETIME	NN

4. Relationships Between Entities

The table below lists all relationships shown on the diagram, together with their cardinality (expressed as "min..max" on each side) and a short description.

Parent Entity	Child Entity	Cardinality	Description
Airport	Flight (departure_airport_id)	1 : 0..N	One airport can be the departure point for many flights; each flight departs from exactly one airport.
Airport	Flight (arrival_airport_id)	1 : 0..N	One airport can be the arrival point for many flights; each flight arrives at exactly one airport.
Airline	Flight	1 : 0..N	One airline operates many flights; each flight belongs to exactly one airline.
Flight	Booking	1 : 0..N	One flight can have many bookings; each booking is made for exactly one flight.
Passenger	Booking	1 : 0..N	One passenger can make many bookings; each booking belongs to exactly one passenger.
Booking	Booking_Change	1 : 0..N	One booking can have several change records; each change record belongs to one booking.
Booking	Boarding_Pass	1 : 0..N	One booking can result in one or more boarding passes (e.g. for connecting flights); each boarding pass belongs to exactly one booking.
Booking	Baggage	1 : 0..N	One booking can include several pieces of baggage; each piece of baggage belongs to one booking.
Baggage	Baggage_Check	1 : 0..N	One piece of baggage can undergo several checks; each check refers to exactly one piece of baggage.
Booking	Baggage_Check	1 : 0..N	A baggage check also references the booking under which the baggage was registered.
Passenger	Baggage_Check	1 : 0..N	A baggage check also references the passenger who owns the baggage.
Passenger	Security_Check	1 : 0..N	One passenger can go through several security checks; each check belongs to exactly one passenger.

Note on participation: in every relationship listed above, the "one" side is mandatory (shown on the diagram as a double bar next to the parent entity), while the "many" side is optional, meaning 0 related records are allowed (for example, a flight may not yet have any bookings). Mandatory participation is marked with a "|" symbol, optional participation with an "o" symbol next to the crow's foot.

5. Notation / Legend

The diagram is drawn using Crow's Foot notation. The symbols at the end of a relationship line, closest to the entity, indicate the cardinality and the participation (optional/mandatory) of that entity in the relationship:

Symbol	Name	Meaning
— —	Exactly one (mandatory "one")	Exactly one related record is required
—o —	Zero or one (optional "one")	A related record may or may not exist, but at most one
—o<—	Zero or many (optional "many")	From 0 to N related records are allowed
— <—	One or many (mandatory "many")	At least one related record is required, with no upper limit
PK	Primary Key	The primary key of the entity
FK	Foreign Key	A reference to the primary key of another entity
UQ	Unique	The attribute value must be unique within the table
NN	Not Null	The attribute is a required field

6. Brief Description of the Solution

The model describes an airport ticket booking system. The central entity is Booking, which links a specific Passenger to a specific Flight. A flight, in turn, is operated by a particular Airline and connects two Airport records - the departure airport and the arrival airport. Several entities depend on Booking: Booking_Change (the history of modifications made to an order), Boarding_Pass (the boarding document), and Baggage (registered luggage). Each piece of baggage may be inspected, producing a Baggage_Check record, and each passenger goes through a security screening, producing a Security_Check record. All relationships in the model are one-to-many (1:N): the "one" side is mandatory and the "many" side is optional, which reflects real-world situations (for example, a flight can exist before it has any bookings at all). There are no many-to-many relationships on the diagram - they are all resolved through associative entities: Booking acts as the link between Passenger and Flight, while Baggage_Check acts as the link between Baggage, Booking and Passenger.